



OT Anomaly & Alarm Monitor

A learning project exploring how operational monitoring data supports OT/ICS security incident investigation.

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• NumPy • Matplotlib | Simulated data project

The Challenge: Detecting Anomalies in Noisy Sensor Data

Continuous Streams

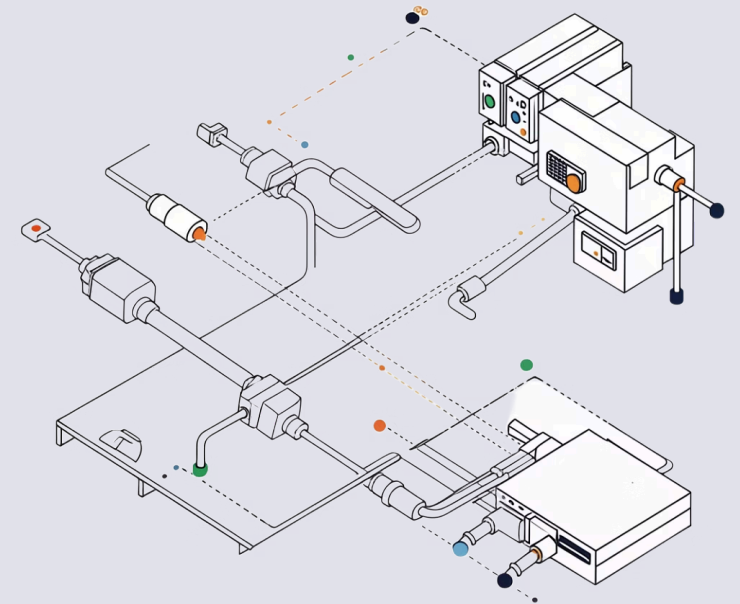
OT systems generate nonstop sensor data.

False Positive Overload

Normal variation can trigger too many alerts.

The Security Distinction

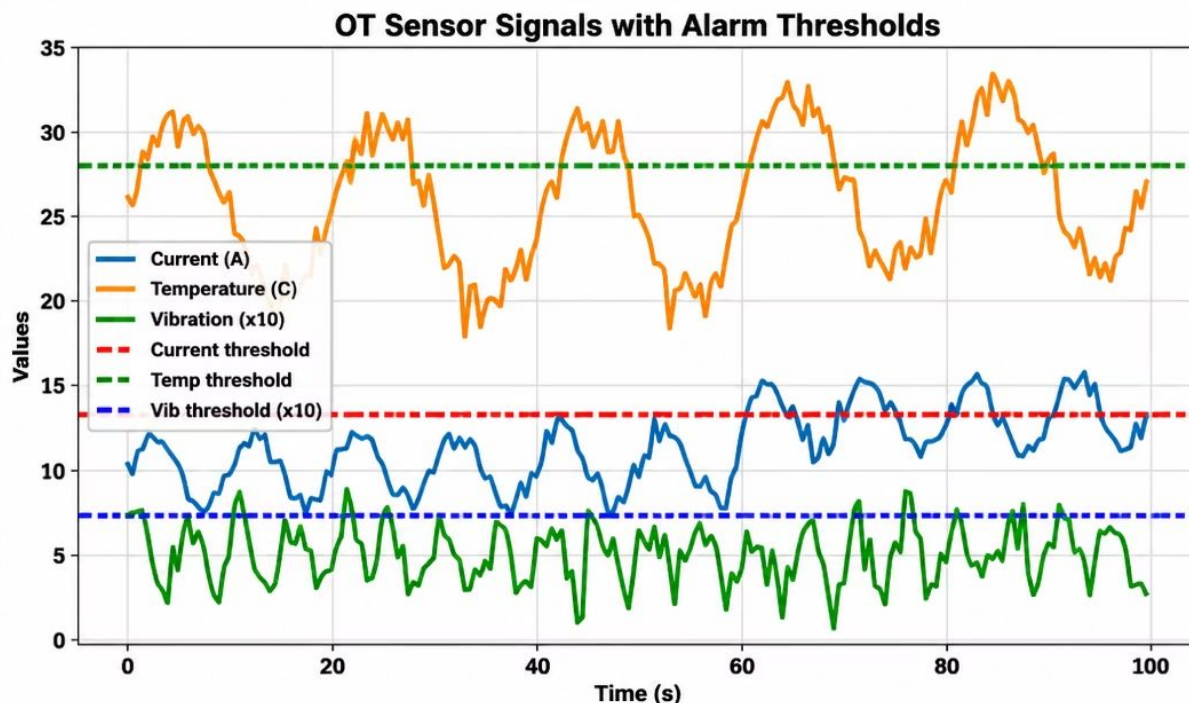
Teams must separate wear from tampering.



#1BB23



Two Detection Approaches



Threshold-Based Alarms

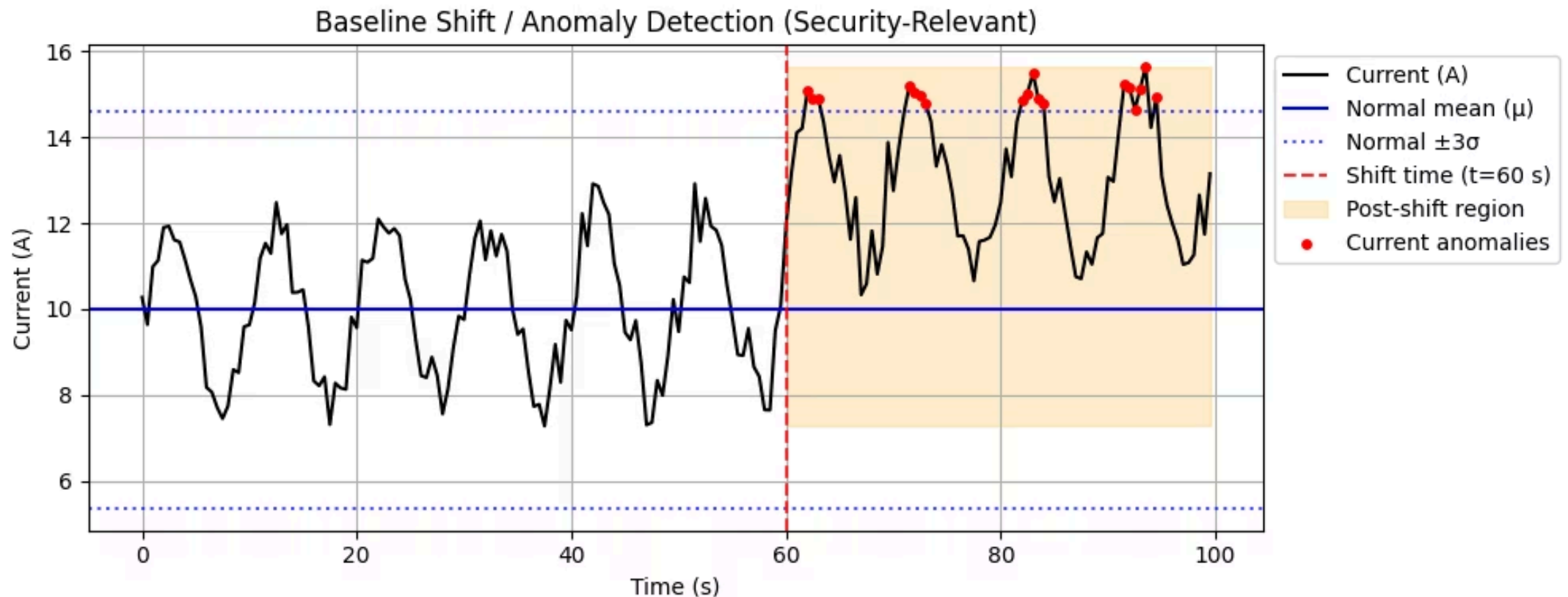
Flag sensor values the moment they exceed predefined limits—delivering an immediate response to extreme or dangerous conditions in the field.

Statistical Anomaly Detection

Identify **baseline shifts**—moments when the normal operating range itself changes. This is the security-critical layer: a shifting baseline can signal equipment drift, control logic changes, or sensor tampering that threshold alarms would never catch.

- Together, these two methods form a layered detection strategy that mirrors real-world OT security monitoring workflows.

What the Data Revealed



18

Current Spikes

Threshold alarms triggered on current excursions

71

Temp Excursions

Temperature threshold breaches detected

9

Baseline Shifts

Security-relevant anomalies where normal operating range shifted

The shift report links alarms to operational events for root cause analysis. Baseline shift detection identified 9 anomalies that threshold-only monitoring would have missed.

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Shift Report (0.0-99.5 s)
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Signal Statistics:
Current (A): mean = 11.21, max = 15.64, std = 2.13
Temperature (C): mean = 25.74, max = 33.28, std = 3.94
Vibration (g): mean = 0.49, max = 0.88, std = 0.17

Alarm Summary:
Current: 41 alarm samples, first at t = 60.5 s
Temperature: 73 alarm samples, first at t = 1.5 s
Vibration: 25 alarm samples, first at t = 0.0 s

Anomaly Summary (statistical vs. normal baseline):
Current anomalies: 18 samples
Temperature anomalies: 0 samples

Security-Relevant Observations:
- Baseline shift configured at t = 60.0 s (simulated setpoint/logic change).
- Current mean before shift: 9.98 A
- Current mean after shift: 13.04 A ( $\Delta = +3.05$  A)
- Temperature mean before shift: 24.95 C
- Temperature mean after shift: 26.93 C ( $\Delta = +1.98$  C)
- Such shifts could indicate:
  * Setpoint change or logic modification in PLC/SCADA
  * Sensor drift or calibration issue
  * Potential cyber incident affecting control parameters
- Recommendation: check change logs, operator actions, and access records around t = 60.0 s.

Conclusion:
- Statistical anomalies detected; behavior after t = 60.0 s differs from initial baseline.
- Correlate with operational events and security logs to determine root cause.
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Why This Matters for OT Security

Incident Response Ready

Can correlate sensor anomalies to security events—the core skill for OT incident investigation and threat hunting.

Real-World Monitoring

Understands how operational data reveals both equipment issues and unauthorized access—critical for SOC and monitoring roles.

Ready for Day One

Demonstrates hands-on capability in anomaly detection, baseline analysis, and security-relevant data interpretation—directly applicable to OT/ICS security positions.